MIL-B-46667A(MU)
10 MAY 1963

SUPERSEDING
MIL-B-46667(Ord)
29 JUNE 1962

MILITARY SPECIFICATION

BURSTER, PROJECTILE, M83
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one projectile burster designated as the M83.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-R-11470 — Radiographic Inspection: Qualification of Equipment, Operators and Procedures.

MIL-R-11471 — Radiographic Inspection of Metals.

MIL-I-45208 — Inspection Requirements, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8860628 — Burster, Projectile, M83.

8863466 — Container, Ammunition Fiber for Burster, Projectile M83 with Loading Accessories.

8863467 — Box, Packing Ammunition for Burster, Projectile M83 with Loading Accessories.

FSC 1320

701044—63—(519)

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PUBLICATIONS

ORDNANCE CORPS

IEL-8860628 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Materials shall be in accordance with applicable drawings and specifications.

3.2 **Burster.** The burster shall comply with all requirements specified on Drawing (dwg.) 8860628 and with all requirements specified in applicable specifications.

3.3 **Cavitation**

3.3.1 *Area from the bottom to one inch above bottom.* There shall be no cavities, cracks or annular rings in the charge in this area.

3.3.2 *Area from top to one inch from bottom of container.* There shall be no single cavity having a projected length of more than one inch or a sum total of cavities of more than $\frac{3}{4}$ square inch and there shall be no more than three transverse cracks $\frac{1}{32}$ inch wide maximum, in this area.

3.3.3 *Separation of charge from base of container.* There shall be no void greater than $\frac{1}{32}$ inch between the charge and the base of the container.

3.3.4 *Porosity.* Porous areas shall be treated as cavities, and shall be subject to the restriction placed upon cavities, except that 80 percent of the projectile length and 80 percent of the projected area shall be considered for acceptance purpose.

3.4 **Workmanship.** All parts shall be free from chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent used.

4. QUALITY ASSURANCE PROVISIONS

4.1 **General quality assurance provisions.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifica-

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tions referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector will verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of projectile burster assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot interfix number. Each lot shall contain:

(a) Composition B4 from not more than one lot.

(b) Metal parts from one manufacturer from not more than one lot interfix number.

(c) Accessories (supplementary charge assembly and projectile burster support) from not more than one lot interfix number.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 *Container, burster, prior to loading.* (See dwg. 8860623 covering a detail of dwg. 8860628.)

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.5)*
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Foreign object in interior	Visual	01001
Minor:	AQL 2.50 percent		
201.	Total length, minimum (min.)	Gage	01002
202.	Thickness through bottom, min	Gage	01003
203.	Outside diameter	Gage	01004
204.	Inside diameter	Gage	01005
205.	Evidence of poor workmanship (see 3.4)	Visual	01006

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4.2.2.2 *Burster projectile.* (See dwg. 8860628.)

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Cushioning pad missing or insecure	Visual	02001
2.	Closure missing	Visual	02002
3.	Burster container cracked	Visual	02003
4.	Explosive charge loose in container (see Note 1)	Manual	02004
5.	"UP" arrow points in wrong direction	Visual	02005
Major:	AQL 0.25 percent		
101.	Failure to chamber gage	Gage	02006
102.	Total length, maximum (without compression of cushioning pad)	Gage	02007
Minor:	AQL 1.50 percent		
201.	Closing disc ruptured or incompletely sealed	Visual	02008
202.	Evidence of poor workmanship (see 3.4)	Visual	02009
203.	Cushioning pad not centered	Visual	02010
204.	Marking misleading or unidentifiable including incorrect color code	Visual	02011

Note 1. The inspection for loose charge shall be conducted by pressing against the charge with the thumb and by inverting the tube and tapping it against the hand. The assembly shall be classed defective if there is any perceptible movement of the charge during this test.

4.2.2.3 *Fiber container prior to sealing.* (See dwg. 8863466.)

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.25 percent		
101.	Burster projectile cannot be removed by hand	Manual	03001
102.	Projectile burster, supplementary charge assembly or projectile burster support missing or from more than one lot interfix	Visual	03002
Minor:	AQL 1.00 percent		
201.	Any packing component missing	Visual	03003
202.	Glue or asphalt on burster projectile	Visual	03004

4.2.2.4 *Fiber container sealed.* (See dwg. 8863466.)

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Cut or damaged through all impregnated layers	Visual	04001
102.	Tape incomplete or badly wrinkled	Visual	04002
103.	Metal end loose or distorted	Visual-Manual	04003
Minor:	AQL 1.50 percent		
201.	Gap between cover and body of container (maximum)	Visual	04004
202.	Contents loose	Manual	04005
203.	Length of tear tab (min.)	Visual	04006
204.	Marking misleading or unidentifiable	Visual	04007

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4.2.2.5 *Sealed wood packing box.* (See dwg. 8863467.)

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Box damaged	Visual	05001
102.	Strapping missing, broken or loose	Manual	05002
103.	Boards loose or nails protruding	Visual-Manual	05003
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled	Visual	05004
202.	Lid improperly assembled	Visual	05005
203.	Marking misleading or unidentifiable	Visual	05006
204.	Contents loose in box	Manual	05007

4.2.3 Testing.

4.2.3.1 *Cavitation (see 3.3).* Cavity, crack or annular ring in the area from the bottom to one inch above the bottom (see 3.3.1) Critical defect—Code No. 06001. Separation of the charge from the base of the container in excess of $\frac{1}{32}$ inch (see 3.3.3) Critical defect—Code No. 06002. Cavitation, cracks or porous areas in excess of that described in 3.3.2 and 3.3.4—Major defects—Code No. 06003. The sampling plan for these defects (Critical and Major) shall be in accordance with Standard MIL-STD-1235 using a CSP-2 plan having an AQL of 0.25 percent, except that when a critical defect is found, 100 percent inspection for i successive defect free units (for both criticals and majors) shall be required immediately instead of continuing on a sample basis. When a major defect is found during the f phase of the plan, inspection shall continue at the same frequency until the homogeneity requirement is fulfilled, or 100 percent inspection is required. Appropriate i, f and k values shall be as specified by the procuring activity. The test shall be performed as specified in 4.3.1.

4.2.4 *Inspection equipment.* Index of Inspection Equipment Lists Number IEL 8860628 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 *Government use of contractor inspection equipment.* The contractor shall as required make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Operation of the equipment and verification of its accuracy shall be supplied by the contractor, as required by the Government in performance of its examination or test.

4.3 Test methods and procedures.

4.3.1 *Determination of cavities.* The sample bursters shall be subjected to X-ray examination. In all exposures the X-ray beam shall form a 90° angle with the longitudinal axis of the projectile. Examination

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of the X-ray negative shall be made to determine compliance with applicable requirements.

Note. Radiographic equipment, operations and procedures shall be qualified in accordance with Specification MIL-R-11740. In addition, the sensitivity of X-ray technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.3.1.1 and 4.3.1.2. In the event an original X-ray picture leaves doubt as to acceptability, the burster shall be re-X-rayed at an angle 90° from that of the original exposure and in the same plane.

4.3.1.1 *Check of sensitivity of X-ray technique.* The sensitivity of X-ray technique shall be determined by means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the burster away from the film (source side). Observation shall be made of the image of the penetrameter on the X-ray film. The sensitivity of X-ray technique shall be considered satisfactory when the image of the penetrameter, including the hole in the center, is as sharply defined on the X-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thicknesses and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see Specification MII-R-11471. The actual thicknesses may be determined from a Radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated in figure 1 of this specification.

4.3.1.2 *Check of photographic density of film.* The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards of radiographic densities be used (see 6.4). Comparison of film with standards may be made visually; however the use of a densitometer is recommended.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The M83 projectile burster shall be preserved and packaged in accordance with dwg. 883466.

5.2 Packing.

5.2.1 Level A. The containers shall be packed in accordance with dwg. 883467.

5.3 Data cards. Data card shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection equipment lists in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design." Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.2 and 6.3.3. Action required by the contractor with respect to "commercial item" inspection equipment is described in paragraph 6.3.4. The contractor will be required to prepare drawings for all the equipment coded as "contractor design" in the number column. The

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code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Ordnance designs.* Ordnance designs are detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an IEL references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirement of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.5 *IEL codes.* The inspection equipment referenced in paragraphs 6.3.1, 6.3.2, 6.3.3, and 6.3.4 will be designated in the IEL by the following codes:

CDOF — Contractor's design responsibility on Ordnance format; ORDM-608-12.

CDCF — Contractor design responsibility on contractor format.

OD — Ordnance design.

ODMU — Ordnance design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.6 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Film density. "American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N.Y. has been found satisfactory as a source of information on film densities.

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Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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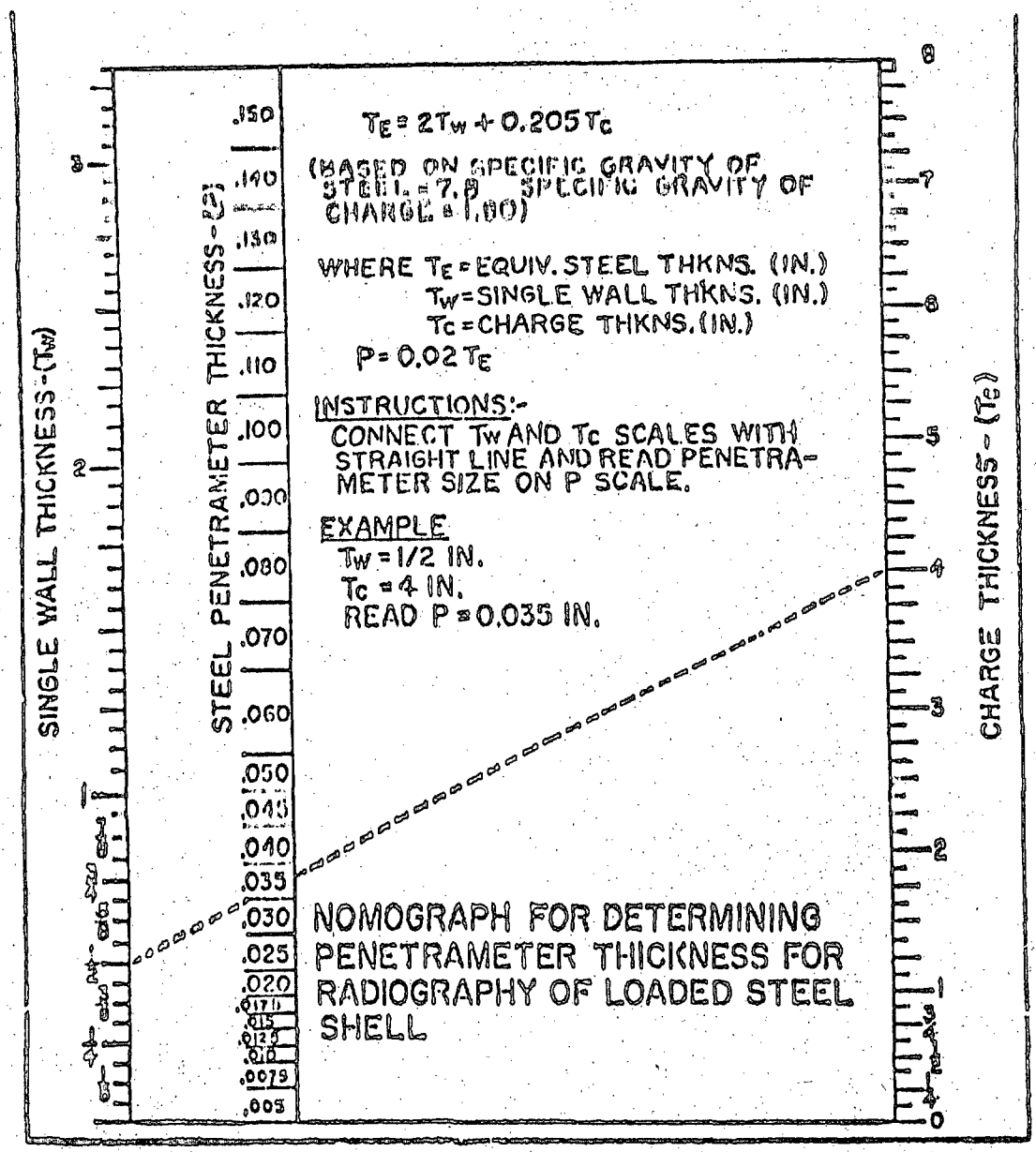


FIGURE 1.

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